

Ci523

13.56MHz contactless reader chip

1. Introduction

This document describes the functional and electrical characteristics of the Ci523 contactless reader/writer.

2. Overview

The Ci523 is a highly integrated contactless reader chip operating at 13.56MHz.

The device supports ISO/IEC 14443 A/B.

No external circuitry is required; the Ci523's internal transmitter can drive the reader antenna in accordance with ISO/IEC 14443.

The A/B card communicates with the transponder. The receiver module provides a powerful and efficient circuit for demodulation and decoding.

ISO/IEC 14443 A/B compliant cards and transponder signals. The digital module processes the complete ISO/IEC 14443 signal.

A/B frames and error detection functions (parity and CRC).

The Ci523 supports contactless communication and provides an SPI (Serial Peripheral Interface) host interface.

It is available in three packages: QFN16, QFN20, and ESOP16.

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3. Parameters and characteristics

- Highly integrated analog circuitry for demodulation and decoding response
- Buffered output driver, requiring minimal external components for antenna connection.
- Supports ISO/IEC 14443 A/B
- The operating distance in reader/writer mode depends on the antenna size and number of turns, with a typical operating distance of 50mm. • Supports ISO/IEC 14443 A/B higher speed communication, up to 848kBd.
- Supports SPI interface with communication rates up to 10 Mbit/s
- 64-byte transmit and receive FIFO buffer
- Flexible interrupt modes
- Low-power hard reset function
- Software power-off mode
- Programmable timer
- Internal oscillator, connected to a 27.12MHz quartz crystal
- 2.3V to 4V supply voltage
- CRC coprocessor



4. Main parameters and indicators

Table 4-1 Limiting Parameters

Working conditions	Minimum value	Maximum value	unit
power supply voltage			
VDD	2.3	4.0	V
temperature			
Operating temperature	-40	+85	°C

Table 4-2 Main Parameter Indicators

Flag parameters		condition		most Small value	class type value	most big value	one Bit
AVDD analog supply	Electric voltage	AVDD = VDD (PVDD) = VDD (TVDD); VSS = 0 V	(1)	2.3	3.3	4.0 V	
VDD (PVDD)	PVDD Power supply			2.3	3.3	4.0 V	
VDD (TVDD)	TVDD Power supply			2.3	3.3	4.0 V	
Ipd	Power outage	AVDD=VDD(PVDD)=VDD(TVDD)= 3.3V					
	flow	Hard power-down; NRSTPD pin low	(2)	-	0.9	2.5 uA	
		Soft power failure;	(2)	-	1.5	1.5 uA	
IPVDD PVDD	Power supply flow	VDD1 pin; PVDD = 3.3 V		-	0.9	1.5 mA	
IDDA analog supply	electric current	VDD4 pin; VDDA = 3.3 V, CommandReg The RcvOff bit of the register is 0.		-	2.9	4 mA	
		VDD4 pin; receiver off; VDDA = 3.3 V. The RcvOff bit of the CommandReg register is set to 1.		-	0.8	1 mA	



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IDD	TVDD	VDD3 pin; TVDD = 3.3 V	(3)	-	25	30 mA
(TVDD)	Power supply					
	flow					

- (1) $VDDA$, VDD ($PVDD$) and VDD ($TVDD$) The voltage must always be kept constant.
- (2) Ipd It is the total current of all power supplies.
- (3) During typical circuit operation, the total current is less than $30mA$.

Table 4-3 Recommended Values

logo	parameter	condition	Minimum value	typical value	maximum value	unit
	AVDD is the analog supply voltage	AVDD = VDD(PVDD) = VDD(TVDD); VSS=0V	2.3	3.3	4.0 V	
VDD(PVD D)	PVDD power supply Pressure		2.3	3.3	4.0 V	
VDD (TVD D)	TVDD power supply Pressure		2.3	3.3	4.0 V	
	Storage temperature	QFN16	-55	-	+125 °	
	Operating temperature	QFN16	-40	-	+85 °	

Note: If the external conditions exceed the rated values of the limit parameters, it will cause permanent damage to the chip.

5. Block Diagram

The analog interface handles the modulation and demodulation of analog signals. The contactless UART handles communication with the host computer.

The protocol requires that the FIFO buffer ensures fast and convenient communication between the host and the contactless UART.

Equipped with an SPI host interface to meet user needs.

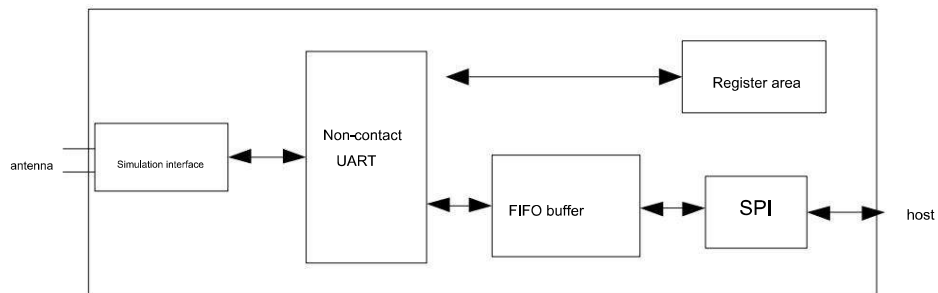


Figure 5.1 Simplified block diagram of Ci523

6. Pin Information

6.1 QFN16 package

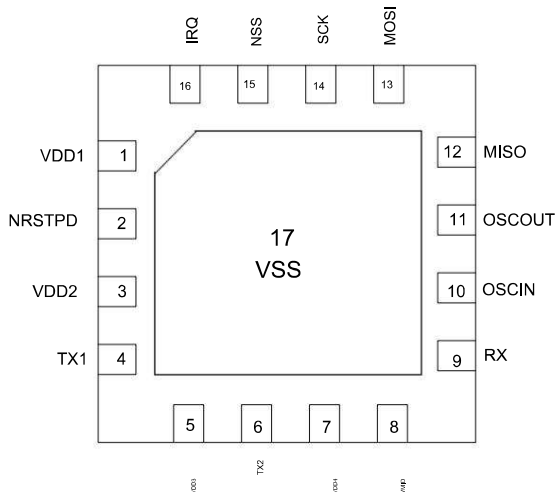


Figure 6-1 Pin configuration diagram (QFN16)

Table 6-1 Pin Description

Pin flag	type [1]		describe
1	VDD1	P	PVDD power supply, pin-side power supply
2	NRSTPD	I	Reset and Power-Down Input Power-down: Low level enables; shuts off internal current, turns off the oscillator, and disconnects the input pin from external connections. Reset: Enable on rising edge
3	VDD2	P	suspended
4	TX1	O	Transmitter 1 outputs modulated carrier signal
5	VDD3	P	Transmitter Power Supply TVDD: Powers the output stages of transmitters 1 and 2.
6	TX2	O	Transmitter 2 outputs modulated carrier signal
7	VDD4	P	Analog power supply AVDD
8	VMD	P	Internal reference voltage
9	RX	I	Radio Frequency Signal Input
10	OSCIN	I	OSCIN I is the input of the inverting amplifier to the crystal oscillator; it is also the input of an externally generated clock.

			(fclk = 27.12 MHz)
11	OSCOUT	O	Output of the inverting amplifier of the crystal oscillator
12	MISO	O	SPI master input, slave output
13	MOSI	I	SPI master output, slave input
14	SCK	I	SPI serial clock input
15	NSS	I	SPI signal input
16	IRQ	O	Interrupt Request Output: Indicates an interrupt event.
17	VSS		The G chip ground is the metal plate at the bottom of the package.

Pin types: I = Input, O = Output, I/O = Input/Output, P = Power Supply, G = Ground

6.2 ESOP16 package

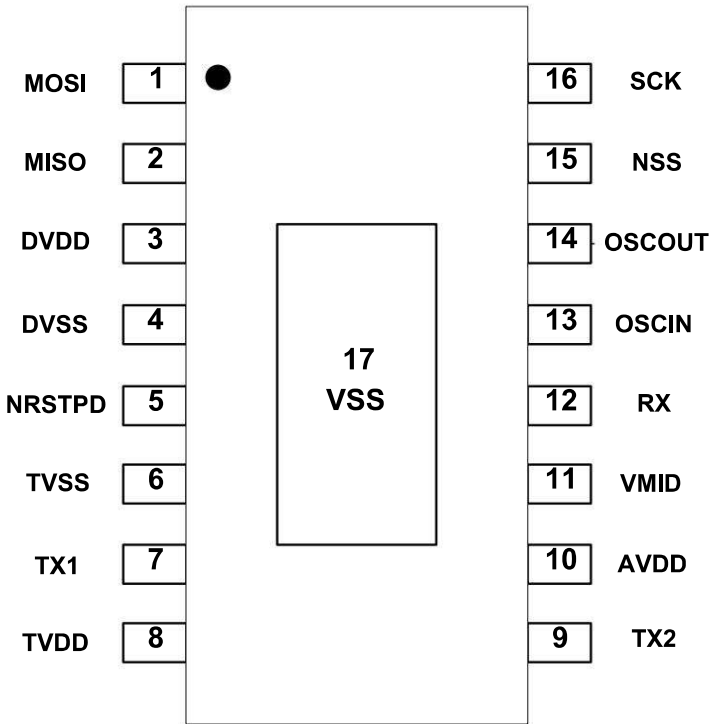


Figure 6-2 Pin Configuration Diagram (ESOP16)

Note: 17 feet VSS It is the metal sheet at the bottom of the package.

Table 6-2 Pin Description



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Pin	flag	type [1]	describe
1		MOSI	I SPI master output, slave input
2		MISO	O SPI master input, slave output
3		DVDD P	Digital Power DVDD
4		DVSS	G grounding
5		NRSTPD I	Reset and Power-Down Input Power-down: Low level enables; shuts off internal current, turns off the oscillator, and disconnects the input pin from external connections. Reset: Enable on rising edge
6		TVSS	G grounding
7		TX1	O Transmitter 1 outputs modulated carrier signal
8		TVDD P	Transmitter Power Supply: TVDD supplies power to the output stages of transmitters 1 and 2.
9		TX2	O Transmitter 2 outputs modulated carrier signal
10		AVDD P	Analog power supply AVDD
11		VMID	P Internal reference voltage
12		RX	I. Radio Frequency Signal Input
13		OSCIN I	I is the input of the inverting amplifier to the crystal oscillator; it is also the input of an externally generated clock. (fclk = 27.12 MHz)
14		OSCOU O	O Output of the inverting amplifier of the crystal oscillator
15		NSS	I SPI signal input
16		SCK	I SPI serial clock input
17		VSS	The G chip ground is the metal plate at the bottom of the package.

Pin types: I = Input, O = Output, I/O = Input/Output, P = Power Supply, G = Ground

6.3 QFN20 package

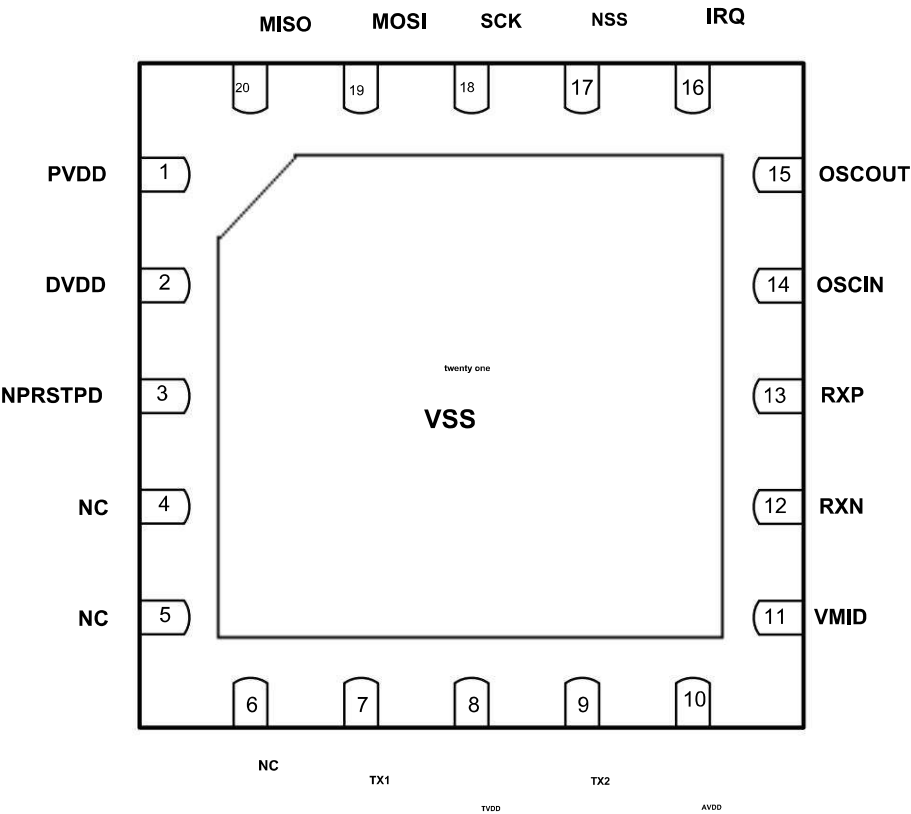


Figure 6-3 Pin configuration diagram (QFN20)

Table 6-3 Pin Description

Pin	flag	type [1]	describe
1	PVDD	P	PVDD power supply, pin-side power supply
2	DVDD	P	DVDD digital power supply
3	NPRSTPD	I	Reset and Power-Down Input Power-down: Low level enables; shuts off internal current, turns off the oscillator, and disconnects the input pin from external connections. Reset: Enable on rising edge
4	NC	— —	
5	NC	— —	
6	NC	— —	



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7	TX1	O	Transmitter 1 outputs modulated carrier signal
8	TVDD P	Transmitter Power Supply:	TVDD supplies power to the output stages of transmitters 1 and 2.
9	TX2	O	Transmitter 2 outputs modulated carrier signal
10	AVDD P	Analog power supply	AVDD
11	VMID	P	Internal reference voltage
12	RXN	I.	Radio Frequency Signal Input
13	RXP	I.	Radio Frequency Signal Input
14	OSCIN	I	I is the input of the inverting amplifier to the crystal oscillator; it is also the input of an externally generated clock. (fclk = 27.12 MHz)
15	OSCON	O	Output of the inverting amplifier of the crystal oscillator
16	IRQ	O	Interrupt Request Output: Indicates an interrupt event.
17	NSS	I	SPI signal input
18	SCK	I	SPI serial clock input
19	MOSI	I	SPI master output, slave input
20	MISO	O	SPI master input, slave output
	VSS		The G chip ground is the metal plate at the bottom of the package.

Pin types: I = Input, O = Output, I/O = Input/Output, P = Power Supply, G = Ground



7. Function Description

The Ci523 transmitter module supports ISO/IEC 14443 A/B at different transmission rates and modulation protocols.

Read/write mode.

Table 7-1 Overview of Communication in ISO/IEC 14443 A Read/Write Mode

Communication direction	Signal type	Transmission rate			
		106kBd	212kBd	424kBd	848kBd
Reader to Card (Ci523 Send Data) Give card)	Reader modulation	100% ASK	100% ASK	100% ASK	100% ASK
	Improved Miller coding				
	Bit length 128 (13.56us)		64 (13.56us)	32 (13.56us)	16 (13.56us)
Card to Reader (Ci523 receiving card sent data)	Card-related modulation	Subcarrier loading adjustment system	Subcarrier loading adjustment system	Subcarrier loading adjustment system	Subcarrier loading adjustment system
	Subcarrier frequency	13.56MHz/16	13.56MHz/16	13.56MHz/16	13.56MHz/16
	Rate				
	Bit-coded Manchester encoding		BPSK	BPSK	BPSK

Table 7. List of Communication-Related Parameters for ISO/IEC 14443B Card Readers

communication direction	signal type	Transmission rate		
		106kBd	212kBd	424kBd
Card reader → Card (Ci523 sent number) According to the card)	Card reader modulation	10% ASK	10% ASK	10% ASK
	Bit Encoding	NRZ -L	NRZ -L	NRZ -L
	Bit length	(128/13.56) μs	(64/13.56) μs	(32/13.56) μs
Card → Card Reader (Ci523 received) (Data from self-card)	Card modulation subcarrier load modulation subcarrier load modulation subcarrier load modulation			
	subcarrier frequency	13.56MHz/16	13.56MHz/16	13.56MHz/16
	Bit Encoding	BPSK	BPSK	BPSK

The Ci523's contactless UART and dedicated peripheral host must comply with the complete ISO/IEC 14443 standard.

A/B Protocol. Figure 7.1 illustrates the data encoding and frame format of the ISO/IEC 14443 A protocol.

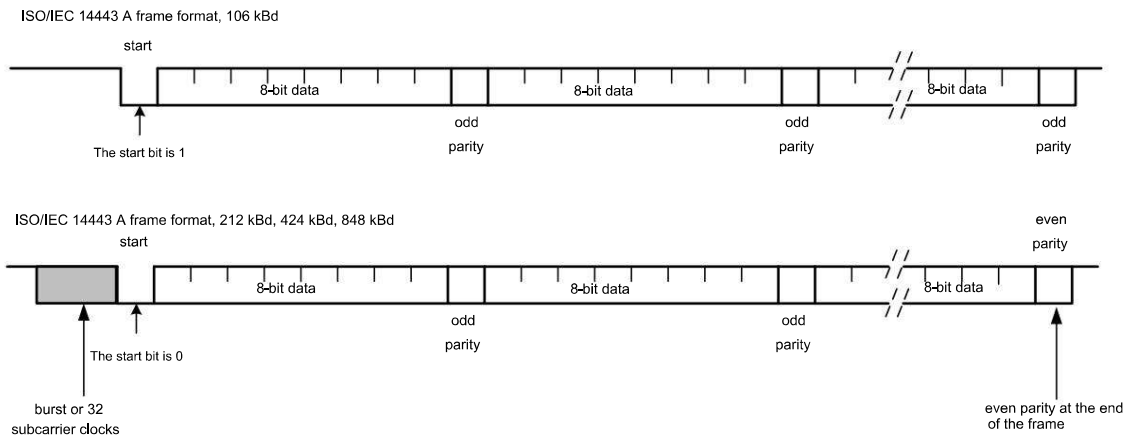


Figure 7.1 Data encoding and frame format of ISO/IEC 14443 A protocol

The internal CRC coprocessor calculates the CRC value according to ISO/IEC 14443 A part 3 and adjusts it according to the transmission rate.

The parity bit is generated automatically. The automatic parity bit generation function is configured via the MfRxReg register.

The ParityDisable bit is turned off.

7.1 Digital Interface

7.1.1 Serial Peripheral Interface

The serial peripheral interface (SPI compatible) supports high-speed communication with the host, up to 10 Mbit/s.

During machine-to-machine communication, the Ci523 acts as a slave device, receiving register configurations from the master and transmitting or receiving RF interface phases.

The interface enables high-speed serial communication between the Ci523 microcontroller and the SPI-compatible interface.

The interface protocol is consistent with the SPI standard.

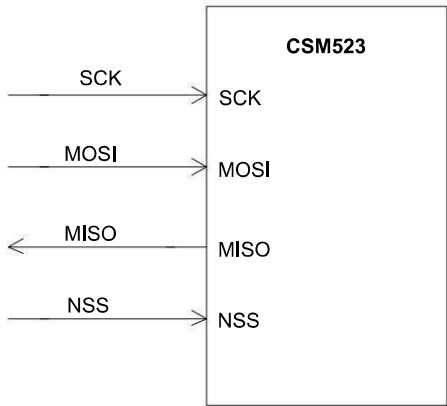


Figure 7.2 SPI connection to the host

When using SPI communication, the Ci523 acts as a slave device, and the SPI clock must be generated by the master device. (This applies to MOSI and MISO.)

All data is MSB first, and the data must remain stable on the rising edge of the clock, but can be modified on the falling edge.

The Ci523 provides data on the falling edge of the clock and remains stable on the rising edge.

7.1.2 SPI Data Reading

The byte order for SPI data reading is shown in the table below. A maximum of n bytes of data can be read, and the first byte sent is defined as follows:

The mode and address are specified. The SPI read timing is shown in Figure 7.3.

Table 7-3 Byte order on MOSI and MISO

Data line	byte 0	Byte 1	2 bytes	to	Byte n	Byte n+1
MOSI	Address 0	Address 1	Address 2	...	Address n	00
MISO	X	Data 0	Data 1	...	Data n-1	Data n

Note: Send the most significant bit first(MSB)

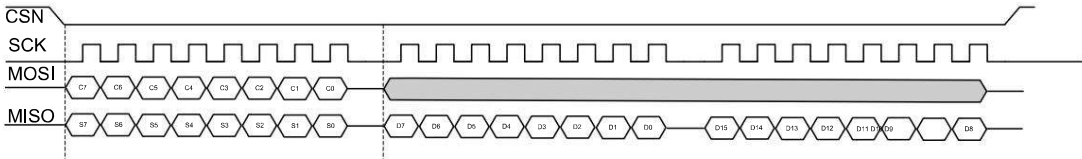


Figure 7.3 SPI Read

7.1.3 Writing Data to SPI

The byte order for writing data via SPI is shown in the table below. Sending the address once allows writing n bytes of data.

The first byte defines the mode and address. The SPI write timing is shown in Figure 7.4.

Table 7-4 Byte order on MOSI and MISO

Data cable	byte 0	Byte 1	2 bytes	to	byte n	byte n+1
MOSI	Address 0	Data 0	Data 1	...	Data n-1	Data n
MISO	X	X	X	...	X	X

Note: Send the most significant bit first(MSB)

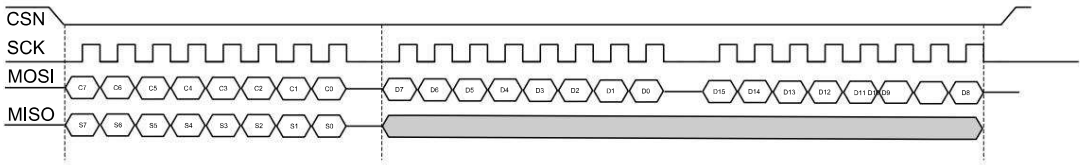


Figure 7.4 SPI writing

7.1.4 Typical SPI Timing

Figure 7.5 shows a typical SPI timing diagram, and Table 7-5 shows typical SPI timing parameters.

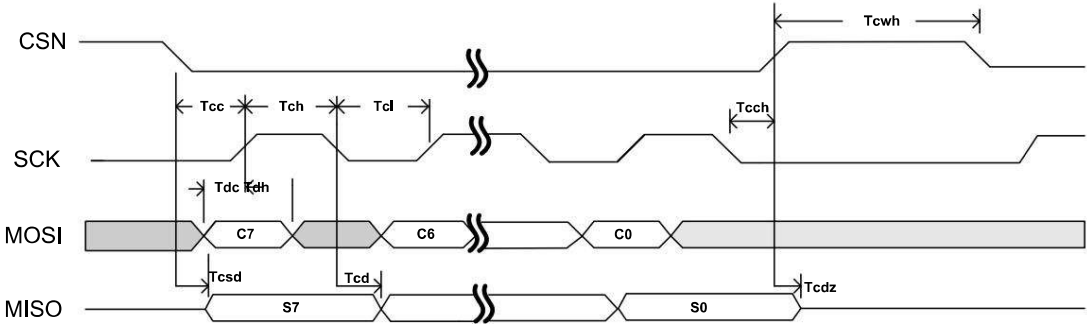


Figure 7.4 Typical SPI timings

Table 7-5 Typical SPI Timing Parameters

Symbol	Parameters	Min	Max	Units
Tdc	Data to SCK Setup	2		ns
Tdh	SCK to Data Hold	2		ns
Tcsd	CSN to Data Valid		42	ns
Tcd	SCK to Data Valid		58	ns
Tcl	SCK Low Time	40		ns
Tch	SCK High Time	40		ns
Fsck	SCK Frequency	0	10	MHz
Tr,Tf	SCK Rise and Fall		100	ns
Tcc	CSN to SCK Setup	2		ns
Tcch	SCK to CSN Hold	2		ns



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Tcwh	CSN Inactive time	50		ns
Tcdz	CSN to Output High Z		42	ns

7.1.5 SPI Address Bytes

The SPI address byte must conform to the following format. The most significant bit (MSB) of the first byte defines the modulus used.

The formula is as follows: the most significant bit (MSB) of the data read back from the Ci523 is logic 1, and the most significant bit of the data written to the Ci523 is...

The MSB must be set to logic 0. Bits 1 to 6 define the address and the least significant bit (LSB) is set to logic 0.

Table 7-6 Address Byte Format

7(MSB)	6:1	0(LSB)
1: Read / 0: Write	address	0

7.2 FIFO Buffer

The Ci523 includes a 64x8-bit FIFO buffer to buffer input between the Ci523 and the host.

Output data stream. The input/output interface is the FIFODataReg register, address 0x09. Data is output by reading and writing to this register.

The register reads and writes to the FIFO buffer. The FIFOLevelReg register indicates the number of bytes stored in the FIFO buffer.

Reading/writing the FIFODataReg register decrements/increments the FIFOLevel value, respectively.

To ensure the host is aware of the status of the FIFO buffer in a timely manner, the FIFO buffer generates two interrupts:

HiAlertIRQ and LoAlertIRQ:

HiAlert is set when the number of bytes in the FIFO buffer satisfies the following equation; if interrupts are enabled, it generates an alarm.

HiAlertIRQ

$$HiAlert = (64 - FIFOLength) - WaterLevel$$

The number of bytes in the FIFO buffer satisfies the following equation, LoAlert is set, and if interrupts are enabled, a value is generated.

LoAlertIRQ

$$LoAlert = FIFOLength - WaterLevel$$

7.3 Interrupt Request System

The Ci523 indicates an interrupt by setting the IRQ bit of the Status1Reg register or activating the IRQ pin.



Table 7-7 Overview of Interrupt Sources

Interrupt flag	Interrupt source	Trigger operation
TimerIRq	Timer unit	Timer counts to 0
TxIRq	transmitter	One data stream transmission ends
CRCIRq	CRC coprocessor	Process all data in the FIFO buffer.
RxIRq	receiver	One data stream receives data
IdleIRq	The ComIrqReg register is used	after a command is executed.
HiAlertIRq	FIFO buffer	The FIFO buffer is about to fill up.
LoAlertIRq	FIFO buffer	The FIFO buffer will be empty.
ErrIRq	Non-contact UART	An error was detected.

7.4 Power Saving Mode

7.4.1 Hard power failure

The chip enters a hard power-down state when the NRSTPD pin is low. After exiting the hard power-down state, all registers...

It becomes the reset value.

7.4.2 Soft power-off

When the PowerDown bit in the CommandReg register is set, the chip immediately enters a soft power-down state and exits.

During a soft power-down, all register values remain unchanged.

7.4.3 Transmitter power failure

The transmitter's power-down mode shuts down the internal antenna driver, effectively disabling the RF field. This is achieved by setting TxControlReg...

Setting either the Tx1RFEn bit or the Tx2RFEn bit in the register to 0 will enter the transmitter power-down mode.

8. Command Set

8.1 Overview

The operation of the Ci523 is determined by a state machine that can execute a series of commands, through the command register.

CommandReg contains command codes that are used to execute commands.

8.2 General features

- In addition to the Transceive command, commands that require a data bit stream (or data byte stream) as input can immediately process any data in the FIFO buffer. When executing the Transceive command, the data is sent by...

StartSend bit of BitFramingReg register to initiate. • Commands requiring a certain

number of parameters will only be initiated if the correct number of parameters are received from the FIFO buffer.

It was only at that time that the process began.

- The FIFO buffer is not automatically cleared when a command starts; writing command parameters to the FIFO buffer is supported.

Count the number of bytes, then start the command.

- The host writes a new command code to the CommandReg register to interrupt the currently executing command, such as the Idle command.

8.3 Command Overview

Table 8-1 Command Overview

Order	Command code	operation
Idle	0000	No operation; cancels the execution of the current command.
Mem	0001	Stores 25 bytes into the FIFO buffer.
Generate RandomID	0010	generates a 10-byte random ID.
CalcCRC	0011	Activate CRC Coprocessor
Transmit	0100	Data in the transmit buffer
NoCmdChange	0111	No command change; used to modify the CommandReg register bits without affecting command execution. Okay, for example, drop the voltage.
Receive	1000	Activate receiver circuit
Transceive	1100	transmits data from the FIFO buffer to the antenna and automatically activates the receiver after transmission is complete.
SoftReset	1111	Reset Ci523

9. Register Map Table

Table 9-1 Register Mapping Table

Address (Hex)	Mnemonic	Bit Type	Reset Value	Description
00h	reserved			00h Reserved
01h	CommandReg			20h Command Register
	reserved	7:6 R/W		The default value is 0.
	RcvOff	5	R/W	1: The analog section of the receiver is turned off.
	PowerDown 4D			1: Enter soft power-off mode 0: This bit is always 1 during the Ci523 startup and wake-up process; 0 indicates... Ci523 is ready Note: If the SoftReset command is activated, this bit cannot be set.
	Command[3:0]	3:0 D		The command is activated based on the command value; reading this register indicates that the command is currently being executed. line command
02h	ComIEEnReg			80h Interrupt Control and Enable Register
	IRqInv	7	R/W	1: The signal on the IRQ pin is related to the IRq value in the StatusReg register. The value of the bit is opposite 0: The signal on the IRQ pin is the same as the value of the IRq bit, and... The IRqPushPull bit of the DivIEEnReg register is used in conjunction with this bit by default. It is assumed that 1 is to ensure that the IRQ pin output level is tri-state.
	TxIEEn	6	R/W	1: Enable transmitter interrupt request (TxIRq bit) to be passed to IRQ. foot
	RxIEEn	5	R/W	1: Enable receiver interrupt requests (RxIRq bit) to be passed to the IRQ. foot
	IdleEn	4	R/W	1: Enable idle interrupt requests (IdleIRq bit) to be passed to the IRQ. foot

	HiAlertIEn	3	R/W		1: Allow alert interrupt requests exceeding the threshold (HiAlertIRq bit) to be transmitted. Passed to IRQ pin
	LoAlertIEn	2	R/W		1: Allow interrupt requests for alerts below the threshold (LoAlertIRq bit) to be transmitted. Passed to IRQ pin
	ErrIEn	1	R/W		1: Enable error interrupt requests (ErrIRq bit) to be passed to the IRQ pin.
	TimerIEn	0	R/W		1: Enable timer interrupt requests (TimerIRq bit) to be passed to IRQ pin
03h	DivIEnReg				00h Interrupt Control and Enable Register
	IRQPushPull 7		R/W		1: IRQ as a standard CMOS pin 0: IRQ is used as an open-drain output pin.
	reserved	6:3	R/W		The default value is 0.
	CRCIEn	2	R/W		1: Enable CRC interrupt requests (via the DivIrqReg register). The CRCIRq bit indicator is passed to the IRQ pin.
	reserved	1:0	-		reserve
04h	ComIrqReg				14h Interrupt Request Flag Register
	Set1	7	W		1: Set the flag bits of the ComIrqReg register. 0: Instructs the flag bit of the ComIrqReg register to be cleared.
	TxIRq	6	D		1: Set the bit immediately after sending the last bit of data.
	RxIRq	5	D		1: Set the bit when the receiver detects the end of a valid data stream. If the RxNoErr bit is set in the RxModeReg register, then RxIRq is set only when valid data exists in the FIFO.
	IdleIRq	4	D		1: If a command terminates (e.g., CommandReg register) (When the device changes from other commands to the Idle command); If an unknown command is initiated, the CommandReg register... The value of Command[3:0] then becomes idle and that position is set; The microcontroller begins executing the Idle command, and this bit is not set.
	HiAlertIRq	3	D		1: When the HiAlertIrq bit of the StatusReg register is set.

					The HiAlertIRq bit, which is the opposite of the HiAlert bit, stores interrupt events. Reset is only possible after the Set1 bit of this register is indicated.
	LoAlertIRq	2D			1: When the HiAlertIrq bit of the StatusReg register is set. The LoAlertIRq bit, which is the opposite of the LoAlert bit, stores interrupt events. Reset is only possible after the Set1 bit of this register is indicated.
	ErrIRq	1	D		1: Any error setting in ErrorReg
	TimerIRq	0	D		1: The count value in TCounterValReg is counted to 0.
05h	DivIrqReg			x0h	Interrupt Request Flag Register
	Set2	7	W		1: Set the flag bits of the DivIrqReg register 0: Clears the flag bits of the DivIrqReg register.
	reserved	6:3	D		reserve
	CRCIRq	2	D		1: The CalcCRC command is activated and all data is processed.
	reserved	1:0	-		reserve
06h	ErrorReg			00h	Error Flag Register
	WrErr	7 R			1: When sending and receiving the last byte at the RF interface Between bytes, the host writes data to the FIFO buffer.
	TempErr	6 R			1: The internal temperature sensor detected overheating, and the antenna driver shut down. Broken
	reserved	5	-		reserve
	BufferOvfl	4 R			1: The host or Ci523 internal state machine (such as the receiver) in the FIFO Writing data to a buffer that is already full
	CollErr	3 R			1: Bit conflict detected The receiver automatically zeros during startup, only at a rate of 106kBd. Effective for anti-collision at bit rates, but ineffective at other rates. 212/424/848kBd
	CRCErr	2 R			1: The RxCRCEn bit in the RxModeReg register is set and the CRC... Calculation failed

					Automatic zeroing during receiver startup
	ParityErr	1	R		1: Parity check failed Automatically zeroed during receiver startup, only at 106kBd. Valid in ISO/IEC 14443 A communication
	ProtocolErr	0	R		1: SOF error The receiver automatically zeros during startup, only at a rate of 106 kBd. efficient
07h	Status1Reg			21h Status Register	
	reserved	7	-		reserve
	CRCOk	6	R		1: CRC result is 0 CRCOk is used when it is undefined during data transmission and reception. The CRCErr bit of the ErrorReg register This bit indicates the CRC coprocessor; it is 0 during the calculation. The calculation is complete and the position is correct.
	CRCReady	5	R		1: CRC calculation complete The CRC coprocessor calculates the CRC only when using the CalcCRC command. Effective
	IRq	4	R		According to the interrupt enable configuration (see ComEnReg and...), DivEnReg indicates the interrupt source of the interrupt request.
	TRunning	3	R		1: The timer unit is running, and when the next clock arrives... The timer will decrement the value of TCounterValReg. Note: In gating mode, the timer is controlled via the TModeReg register. After enabling TGate[1:0] of the register, the TRunning bit is set. Unaffected by gating signals
	reserved	2	-		reserve
	HiAlert	1	R		1: The number of bytes in the FIFO buffer satisfies the following equation $HiAlert = (64 - FIFOLength) \gg WaterLevel$



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					example: FIFO length = 60, WaterLevel = 4 -> HiAlert = 1 FIFO length = 59, WaterLevel = 4 -> HiAlert = 0
	LoAlert	0 R			1: The number of bytes in the FIFO buffer satisfies the following equation LoAlert = FIFOLength÷WaterLevel example: FIFO length = 4, WaterLevel = 4 -> LoAlert = 1 FIFO length = 5, WaterLevel = 4 -> LoAlert = 0
08h	Status2Reg			00h	Status Register
	TempSensClear	7	R/W		1. When the temperature is below the alarm threshold of 125y, the temperature error is cleared. remove
	reserved	6:3	-		reserve
	ModemState[2:0] 2:0 R				Instructing the state machine of the transmitter and receiver 000: Idle 001: Waiting for the StartSend bit of the BitFramingReg register to be set. 010: Send Wait (TxWait), if TModeReg register If the TxWaitRF bit is set, the device remains in the TxWait state until the signal is emitted. The frequency field exists, and the minimum TxWait time is determined by TxWaitReg. Register definition 011: Sending 100: Receive Wait (RxWait), if the TModeReg register... If the TxWaitRF bit is set, it will be in the RxWait state until the emission occurs. The frequency field exists, and the minimum RxWait time is determined by RxWaitReg. Register definition 101: Waiting for data 110: Receiving
09h	FIFODataReg			xxh	FIFO buffer input/output register

	FIFOData[7:0]	7:0 D			Internal 64-byte FIFO buffer for data input/output ports A FIFO buffer serves as a parallel input/output buffer for all data streams. Input/output converter
0Ah	FIFOLevelReg			00h	FIFO Buffer Byte Count Indicator Register
	FlushBuffer	7 W			1: Immediately clear the internal FIFO read/write pointers and ErrorReg. BufferOvfl bit of register Reading this bit always returns 0
	FIFOLevel[6:0]	6:0 R			Indicates the number of bytes stored in the FIFO buffer. Read/write the FIFODataReg register to decrement/increment FIFOLevel respectively. value
0Bh	WaterLevelReg			08h	FIFO overflow/underflow warning threshold register
	reserved	7:6	-		reserve
	WaterLevel[5:0]	6:0 R/W			Define warning values to indicate FIFO buffer overflow or underflow. If the number of remaining bytes in the FIFO buffer is less than or equal to the definition The WaterLevel value, the HiAlert value in the Status1Reg register. Position If the FIFO buffer length is less than or equal to the defined... WaterLevel value, LoAlert position in Status1Reg register. Bit
0Ch	ControlReg			10h	Control Register
	TStopNow	7 W			1: Stop the timer immediately Reading this bit always returns 0
	TStartNow	6 W			1: Timer starts immediately Reading this bit always returns 0
	reserved	5:3	-		reserve
	RxLastBits[2:0]	2:0 R			Indicates the number of valid bits in the last received byte. If the value is 000b, the entire byte is valid.

0Dh Bit	FramingReg			00h Bit-oriented	Frame adjustment register
	StartSend	7 W			1: Start transmitting data Only valid when used with the Transceive command.
	RxAlign[2:0] 6:4 R/W				For receiving bit-oriented frames: defines the first received frame. The location where the bit is stored in the FIFO buffer, for example: 0: The received LSB is stored in bit 0, and the second bit is stored in bit 1. 1: The received LSB is stored in bit 1, and the second bit is stored in bit 2. 7: The received LSB is stored in bit 7, and the second bit is stored immediately following. The next byte of bit 0 These bits are only used for bit anti-collision functions at 106kBd; others... All in mode are 0
	reserved	3	-		reserve
	TxLastBits[2:0]	2:0 R/W			For frame-oriented transmission: defines the last byte to be transmitted. number of digits 000b indicates that all bits of the last byte should be sent.
0Eh	CollReg			xxh RF interface	First bit collision detection related definition register
	ValuesAfterColl	7	R/W		0: All received bits are cleared after a collision. Used only at 106kBd for collision prevention; otherwise... It should be set to 1
	reserved	6	-		reserve
	CollPosNotValid	5	R		1: No collision detected or collision bits are in CollPos[4:0]. Outside the scope
	CollPos[4:0]	4:0 R			It shows the position of the bit where the first collision was detected in the received frame. Only display the data bit positions, for example: 00h: Indicates that the conflict bit is at the start bit. 01h: Indicates that the conflict bit is in the 1st position. 08h: Indicates the conflict bit is in the 8th position.

					These bits are only valid when the CollPosNotValid bit is 0.
0Fh	reserved				reserve
10h	reserved			00h Reserved	
11h	ModeReg			3Fh Transceiver related mode configuration register	
	MSBFirst	7	R/W		1: The CRC coprocessor calculates the CRC starting from the MSB. Bits CRCResultMSB[7:0] of the CRCResultReg register The CRCResultLSB[7:0] bits are matched in binary bit order. The opposite Note: This bit is ignored during radio frequency communication.
	reserved	6	-		reserve
	TXWaitRF	5	R/W		1: The transmitter only starts after the radio frequency field is generated.
	reserved	4:2 R/W			reserve
	CRCPreset	1:0 R/W			Define the default values for the CRC coprocessor to execute the CalcCRC command. Note: During any communication, the default value is based on RxModeReg and The definition bits of the TxModeReg register are automatically selected. 00:0000h 01:6363h 10: A671h 11:FFFFh
12h	TxModeReg			00h	Define the data rate when sending.
	TxCRCEn	7	R/W		1: Enable CRC generation during data transmission Note: Can only be set to 0 at a rate of 106kBd.
	TxSpeed[2:0] 6:4 D				Define the bit rate for transmitting data, up to a maximum of 848 kBd. 000:106kBd 001:212kBd 010:424kBd 011:848kBd

					100-111: Retain
	InvMod	3	R/W		1: Modulation inversion of transmitted data
	TxFraming	1:0 R/W			00: ISO/IEC 14443 A 01: Retain 10: Retain 11: ISO/IEC 14443 B
13h	RxModeReg			00h defines	the data rate during reception.
	RxCRCEn	7	R/W		1: Enable CRC generation during data reception Note: Can only be set to 0 at a rate of 106kBd.
	RxSpeed[2:0]	6:4 D			Define the bit rate of received data, up to 848 kBd. 000:106kBd 001:212kBd 010:424kBd 011:848kBd 100-111: Retain
	RxNoErr	3	R/W		1: Ignore an invalid data stream (receive less than 4 bits), receive. The device is still working.
	RxMultiple	2	R/W		0: The receiver closes after receiving a data stream. 1: Capable of receiving multiple data streams
	RxFraming	1:0 R/W			00: ISO/IEC 14443 A 01: Retain 10: Retain 11: ISO/IEC 14443 B
14h	TxControlReg			80h	Control antenna drive pins TX1 and TX2
	InvTx2RFOn	7	R/W		1: When TX2 is enabled, the output signal of the TX2 pin is inverted.
	InvTx1RFOn	6	R/W		1: When TX1 is enabled, the output signal of the TX1 pin is inverted.
	InvTx2RFOff	5	R/W		1: When TX2 is disabled, the output signal of the TX2 pin is inverted.

	InvTx1RFOff	4	R/W		1: When TX1 is disabled, the output signal of the TX1 pin is inverted.
	Tx2CW	3	R/W		1: The TX2 pin output is a continuous, unmodulated 13.56MHz. carrier 0: Tx2CW bit enable modulation of 13.56MHz carrier
	reserved	2	-		reserve
	Tx2RFE _n	1	R/W		1: The TX2 pin transmits data at a frequency of 13.56MHz. carrier
	Tx1RFE _n	0	R/W		1: The TX1 pin transmits data at a frequency of 13.56MHz. carrier
15h	TxASKReg			0 0h	Modulation configuration for controlling transmission
	reserved	7	-		reserve
	Force100ASK	6	R/W		1: Independent of ModGsPReg register configuration, forced 100% ASK modulation
	reserved	5:0	-		reserve
16h	TxSelReg			10h	Select internal signals for the analog module
	reserved	7:6	-		reserve
	DriverSel[1:0]	5:4 R/W			Select the inputs for drives TX1 and TX2. 00: Tri-state. If DriverSel[1:0] is set to tri-state mode, In soft power-down mode, the driver only has a tri-state mode. 01: Modulated signal (envelope) from the internal encoder, Miller pulse punch code 10: Retain 11: High level, high level depends on InvTx1RFO _n /InvTx1RFO _{ff} and Setting the InvTx2RFO _n /InvTx2RFO _{ff} bits
	reserved	3:0 R/W			reserve

17h	RxSelReg			84h	Select internal receiver configuration
	UARTSel[1:0]	7:6 R/W			Select contactless UART input 00: Constant low level 01: Retain 10: Default, modulation signal of internal analog module 11: Retain
	RxWait[5:0] 5:0 R/W				After the data is sent, the receiver will turn on after a delay of RxWait clock cycles. During this "frame protection time," any signal on the RX pin... The number was ignored. The Receive command ignores this parameter. All other commands, such as Transceive, use this parameter. The counter starts counting immediately after the external RF field is turned on.
18h RxThresholdReg					Select the threshold of the bit decoder
	MinLevel[3:0]	7:4 R/W			Define the minimum input signal strength that the decoder can accept. Signals with a strength below this value are invalid.
	reserved	7:4	-		reserve
	CollLevel[2:0]	3:0 R/W			Define the minimum signal strength at the decoder input, Manchester encoding. The weak half of the signal must reach this strength to be generated from the stronger half. A bit conflict related to bit amplitude
19h	DemodReg				Demodulator setting register
	AddIQ[1:0]	7:6 R/W			Defines the use of I and Q channels during reception. Note: The FixIQ bit must be set to 0 to enable the following configurations: 00: Choose a stronger channel 01: In communication, select the stronger channel and freeze the selected channel. 10: Retain 11: Retain
	FixIQ	5	R/W		1: If AddIQ[1:0] is set to X0b, the receiver will always be on channel I;

					If AddIQ[1:0] is set to X1b, the receive channel is fixed to Q.
	TPrescalEven	4	R/W		Set frequency division mode
	TauRcv[1:0] 3:2 R/W				Changing the time constant of the internal PLL during data reception
	TauSync[1:0] 1:0 R/W				Changing the time constant of the internal PLL during a burst
1Ah	reserved			00h	reserve
1Bh	reserved			00h	reserve
1Ch	MfTxReg			62h	Control some communication sending parameters
	reserved	7:2	-		reserve
	TxWait	1:0 R/W			Define additional response time
1Dh	MfRxReg			00h	Control some communication receiving parameters
	reserved	7:5	-		reserve
	ParityDisable 4		R/W		1: Generate the parity bits required for transmission, and simultaneously disable reception. The parity check receives the parity bit as a data bit. To handle
	reserved	3:0	-		reserve
1Eh	TypeBReg			00h	Configure ISO/IEC 14443 B protocol
	RxSOFReq 7				When set to 1, data streams without SOF are not received; when cleared to 0, Receives data streams with and without SOF. SOF will not be written. Data streams without SOF are entered into the FIFO, SOF data will not be written to the FIFO.
	RxEofReq 6				When set to 1, data streams without EOF are not accepted, and no data stream ends. The data stream at EOF will cause ProtocolErr; after clearing to 0, the receiving... Data streams with and without EOF. EOF data will not be written. FIFO
	reserved	5			
	EOFSOFWidth	4			If set to 1 and EOFSOFAdjust is 0, SOF and EOF Take the maximum length defined in the ISO 14443B protocol; If it is cleared to zero and EOFSOFAdjust is 0, SOF and EOF take...

					The minimum length defined in the ISO 14443B protocol; If it is set to 1 and EOFSOFAadjust is 1, then: SOF low-level time: $SOFlow = (11etu - 8cycles) / fc$ SOF high-level time: $SOFhigh = (2etu + 8cycles) / fc$ EOF low-level time: $EOFlow = (11etu - 8cycles) / fc$ Where etu is the duration of 1 bit, and cycle is 1 hour. Clock period, fc is the carrier frequency. If set to 0 and EOFSOFAadjust is 1, the system Behavior that does not comply with ISO standards
	NoTxSOF	3			When set to 1, no SOF is generated when transmitting data.
	NoTxEOF	2			When set to 1, the transmitted data does not generate an EOF (Extended Message of Failure).
	TxEgt	1:0			Define the length of the inter-character protection time (EGT). 00: 0bit 01: 1 bit 10: 2 bits 11: 3 bits
1Fh	reserved			EBh	reserve
20h	reserved			00h	reserve
21h	CRCResultReg (higher bits)			MSB of FFh CRC calculation result	
	CRCResultMSB [7:0]	7:0 R			This indicates that the value of the high byte of the CRCResultReg register is only... The Status1Reg register is valid when the CRCReady bit is 1.
22h	CRCResultReg (lower bits)			FFh	LSB of CRC calculation result
	CRCResultLSB [7:0]	7:0 R/W			This indicates that the value of the low byte of the CRCResultReg register is only... The Status1Reg register is valid when the CRCReady bit is 1.

23h	reserved			88h	reserve
24h	ModWidthReg			26h	Configure modulation width
	ModWidth[7:0]	7:0 R/W			The width of the Miller modulation is defined as a multiple of the carrier frequency. The maximum value of (ModWidth+1/Fclk) is half the bit period.
25h	reserved			87h	reserve
26h	RFCfgReg			48h	Configure receiver gain
	reserved	7	-		reserve
	RxGain[2:0]	6:4 R/W			Define the receiver's signal voltage gain parameters 000: 18dB 001: 23dB 010: 18dB 011: 23dB 100: 33dB 101: 38dB 110: 43dB 111: 48dB
	reserved	3:0	-		reserve
27h	GsNReg			88h	Define the conductance of the n-drivers for pins TX1 and TX2.
	CWGsn[3:0]	7:4 R/W			In the unmodulated state, define the N-drive conductance value of the output; it can be... Used to adjust the output power, thereby adjusting the current and working distance. Only effective when antenna drive is enabled; highest bit in soft power-down mode It must be 1. Note: The conductivity values are binary weighted.
	ModGsN[3:0]	3:0 R/W			In the unmodulated state, the N-drive conductance value of the output is defined to be used as... Adjust modulation coefficients; only effective when antenna drive is on; soft mode. In electrical mode, the highest bit must be 1. Note: The conductivity values are binary weighted.

28h	CWGsPReg			20h	Define the conductance of the p-driver during the no-modulation period.
	reserved	7:6	-		reserve
	CWGsP[5:0] 5:0 R/W				Defining the conductance value of the output P-driver can be used to adjust the output. Power, thereby adjusting current and operating distance; soft power-down mode The highest bit must be 1. Note: The conductivity values are binary weighted.
29h	ModGsPReg			20h	Define the conductance of the p-driver during modulation.
	reserved	7:6	-		reserve
	ModGsP[5:0] 5:0 R/W				When modulation is present, the output P-driven conductance value is defined, which can be... Used to adjust the modulation coefficient; if Force100ASK is 1, The ModGsP value is invalid; the minimum CWGsP value in soft power-down mode. The highest bit must be 1. Note: The conductivity values are binary weighted.
2Ah	TModeReg			00h	Define timer settings
	TAuto	7	R/W		1: The timer operates at all rates and in all communication modes. Automatically start after sending is complete 0: The timer is not affected by the protocol.
	reserved	6:5			reserve
	TAutoRestart 4		R /W		1: The timer does not count to 0; it is reloaded from the 16-bit timer. The value starts counting again. 0: The timer counts to 0 and the ComIrqReg register... TimerIrq is set
	TPrescaler_Hi	3:0 R/W			Defines the high 4 bits of the frequency divider.
2Bh	TPrescalerReg			00h	defines the lower 8 bits of the frequency divider.
	TPrescaler_Lo	7:0 R/W			Defines the lower 8 bits of the frequency divider.



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2Ch	TReloadReg (higher bits)			00h defines	the high eight bits of the auto-reload value for the 16-bit counter.
2Dh	TReloadReg (lower bits)			00h defines	the lower eight bits of the auto-reload value for the 16-bit counter.
2Eh	TCounterValReg (higher bits)			xxh	The high eight bits of the timer value
2Fh	TCounterValReg (lower bits)			xxh	The lower 8 bits of the timer value
30h ~35h	RFT			00h	
36h	AutoTestReg			40h	
	reserved	7:5	-		reserve
	EOFSOFAdjust	4	R/W		If EOFSOFWidth is set to 1 and EOFSOFAdjust The value is 0, and SOF and EOF are taken from the values defined in the ISO 14443B protocol. Maximum length; If EOFSOFWidth is cleared to zero and EOFSOFAdjust is 0, SOF and EOF are taken from the minimum length defined in the ISO 14443B protocol. Spend; If EOFSOFWidth is set to 1 and EOFSOFAdjust If the value is 1, then: SOF low-level time: $SOF_{low} = (11et_u - 8cycles) / f_c$ SOF high-level time: $SOF_{high} = (2et_u + 8cycles) / f_c$ EOF low-level time: $EOF_{low} = (11et_u - 8cycles)$ /f_c, Where et_u is the duration of 1 bit and cycle is 1 clock cycle. During the period, f_c is the carrier frequency. If EOFSOFWidth is set to...



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					With 0 and EOFSoFAdjust being 1, the system behavior does not conform to ISO. standard
	reserved	3:0	-		reserve
37h	Version		R	B2h version number	

Note: *R/W* Reading and writing; *D* :dynamic; *R* Read-only; *W* Write only;

10 Typical Application Diagrams

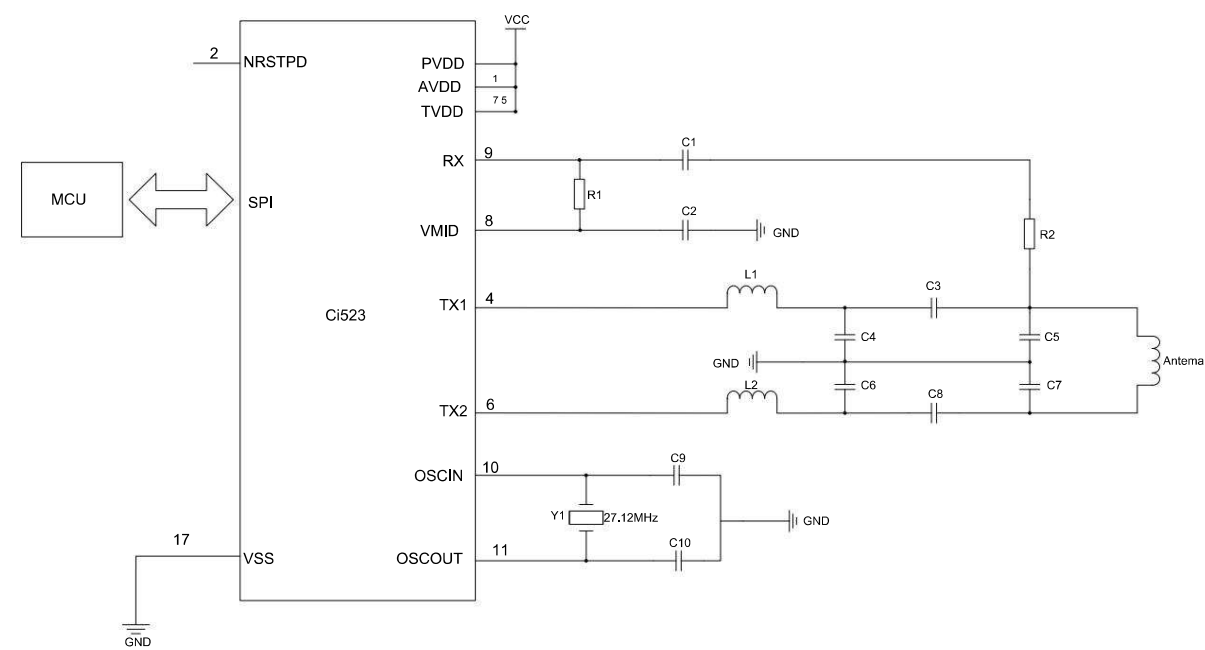


Figure 10.1. Typical Application Diagram-1

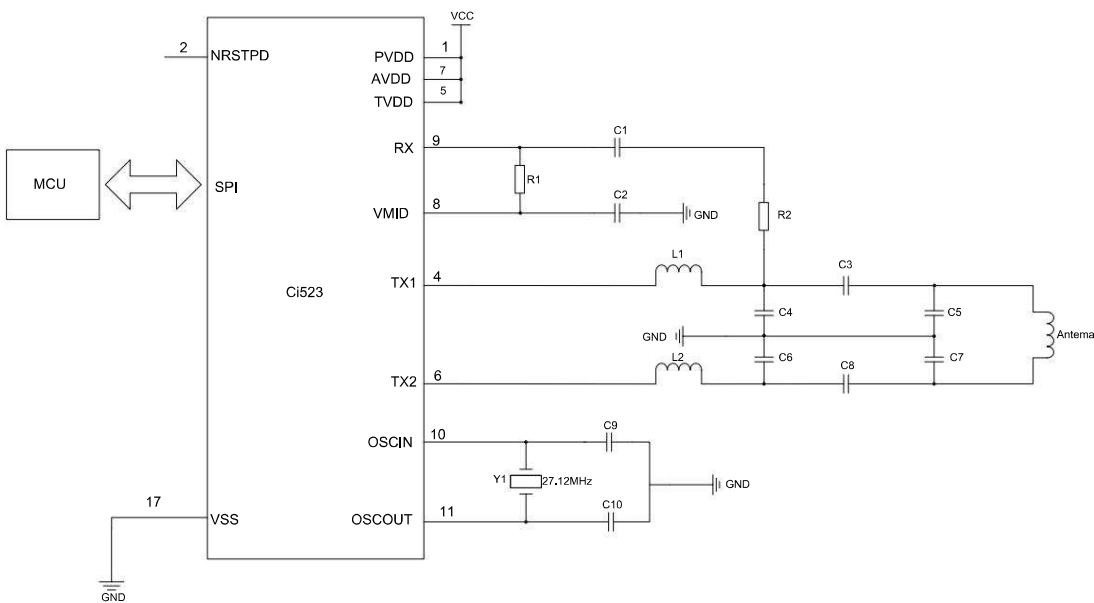


Figure 10.2 Typical Application Diagram - 2

Note: When using a button battery, it is recommended to add a power supply to the power supply section. 100uF Large capacitor;

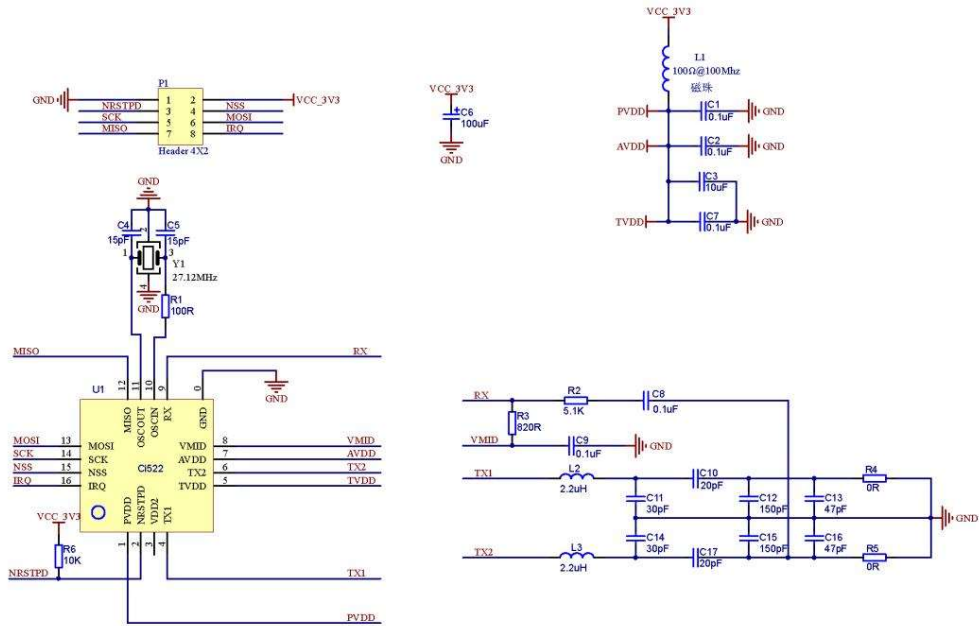


Figure 10.3 Typical application schematic (QFN-16 package)

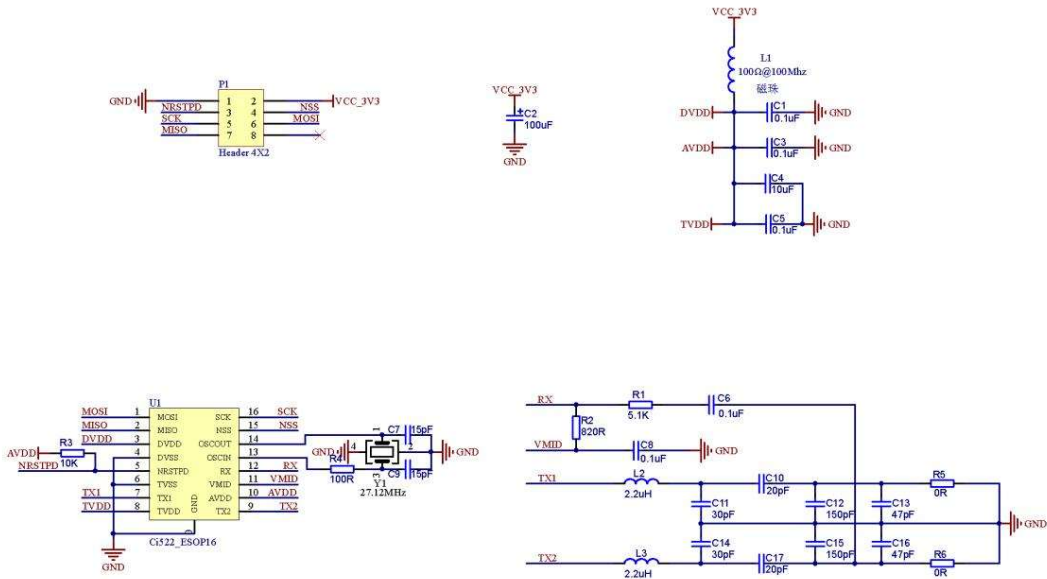


Figure 10.4 Typical application schematic (ESOP-16 package)

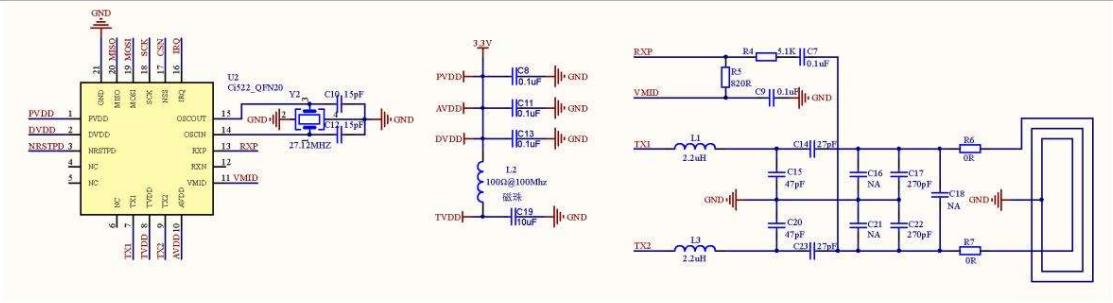


Figure 10.5 Typical application schematic (QFN-20 package)

11. Package size

11.1 QFN16 package

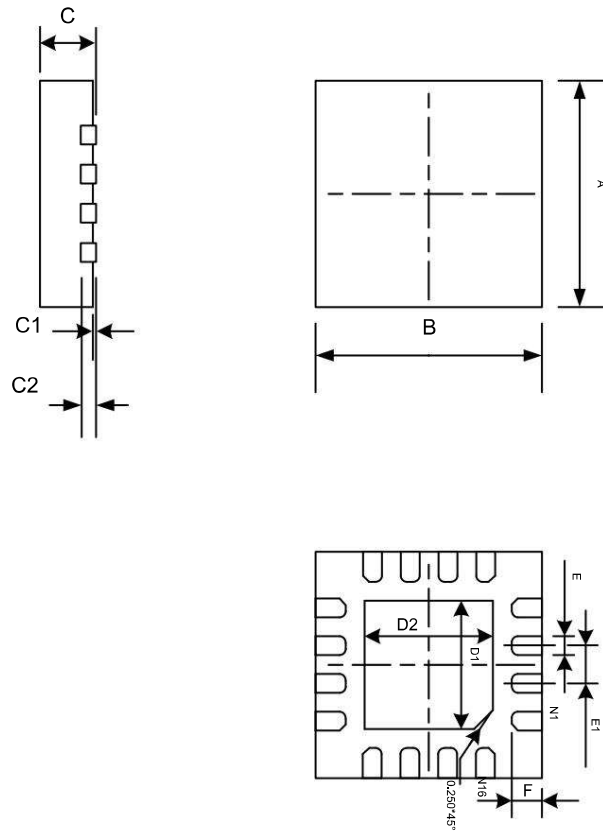


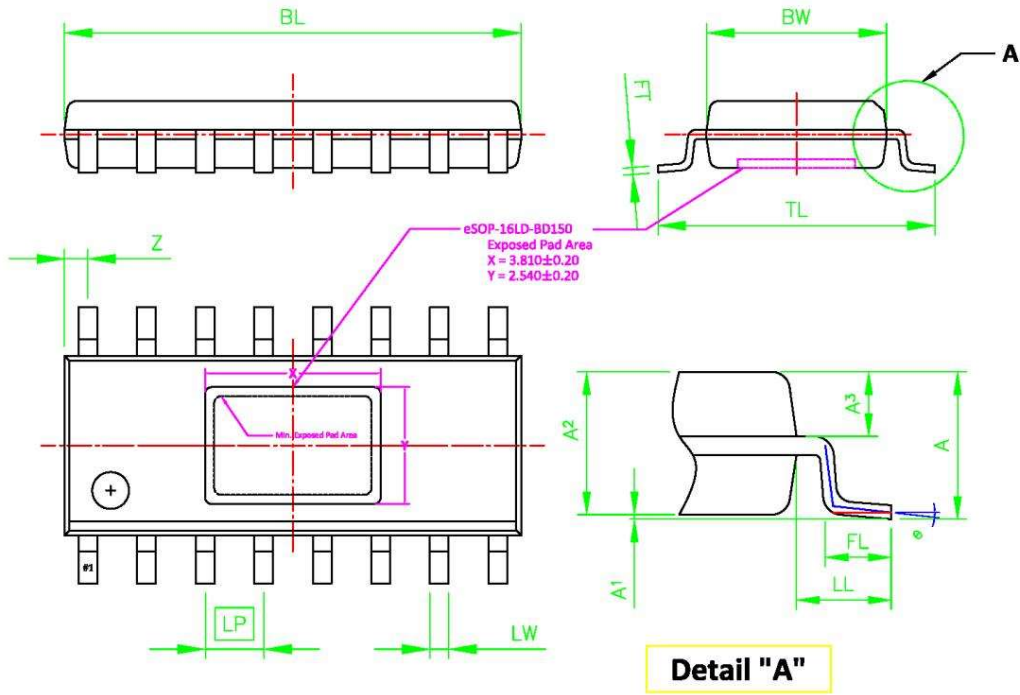
Figure 11.1 Chip package (QFN16)

Table 11-1 Package Dimensions

size Label	min	max	size Label	min	max
A	3.0 ± 0.1		D1	1.70TYP	
B	3.0 ± 0.1		D2	1.70TYP	
C	0.70	0.80	E	0.250TYP	
C1	0~0.050		E1	0.500TYP	
C2	0.203TYP		F	0.400TYP	

Unit: mm

11.2 ESOP16 package



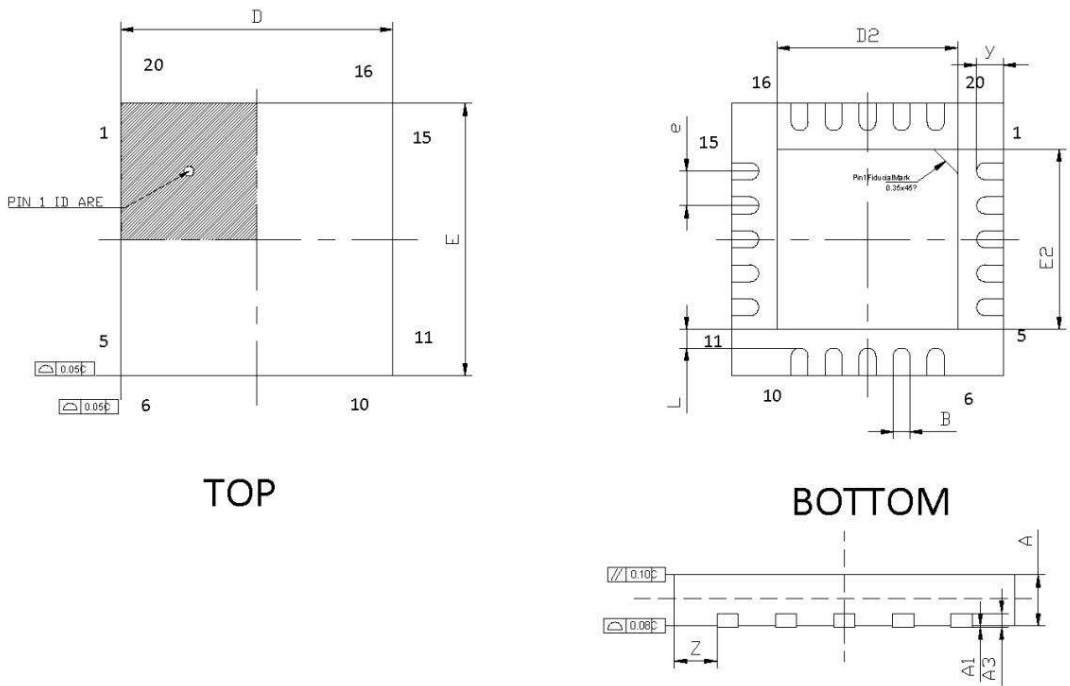
Dimensions

Unit	BL	BW	FT	TL	LP	LW	A	A1	A2	A3	LL	FL	θ	Z
mm	10.00 (9.90) 9.80	4.00 (3.90) 3.80	0.211 (0.203) 0.195	6.10 (6.00) 5.90	1.295 (1.270) 1.245	0.425 (0.400) 0.375	1.75 1.60	0.25 0.05	1.55 1.35	0.623 BSC	1.05 BSC	0.80 0.50	8 0	(0.50)

- Notes:
1. All Dimensions are in Millimeters.
 2. Dimensions Do Not include Burrs, Mold Flash, and Tie-bar Extrusions.
 3. Dimensions(LW) Do Not include Plating Thickness.
 4. JEDEC References : MS-012
 5. Mold Flash should Not be over 0.200mm per each side on the Exposed Pad.

Figure 11.2 Chip package (ESOP16)

11.3 QFN20 package



Dimensions

Unit	D	E	D2	E2	A	A1	A3	B	e	K	L	y	Z
mm	3.025 (3.00) 2.975	3.025 (3.00) 2.975	1.65 (1.6) 1.55	1.65 (1.6) 1.55	0.80 (0.75) 0.70	0.05 (0.02) 0.00	0.203 REF	0.30 (0.25) 0.20	0.40 BSC	-	0.33 (0.28) 0.23	0.40 REF	0.655 REF

Notes

- 1. All Dimensions are in Millimeters.
- 2. Dimensions Do Not Include Burrs, Mold Flash, and Tie-bar Extrusions.

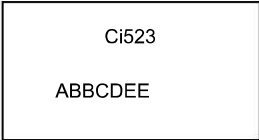
Figure 11.3 Chip package (QFN20)

12. Version Information

Version	modification date	Modifications
V1.3	2021/12/02	Modified contact information
V1.4		Order information modified on 2022/10/24
V1.5		Added QFN16 schematic, ESOP16 schematic and package dimensions on 2022/11/08
V1.6	2022/12/23	Updated ESOP16 package pinout diagram
V1.7		Updated maximum operating voltage and temperature parameters on 2023/02/06
V1.8		Added QFN20 pinout and package diagram on 2023/03/22
V1.9	2023/04/14	Added typical application schematic diagram for QFN20
V2.0	July 21, 2023:	Modified units for some parameters.
V2.1	2023/11/13	Some format optimizations

13. Order Information

Package markings



Ci523: Chip Code

A: Encapsulate the year code for the date; 5 represents 2020.

BB: Processing order issuance week log, for example, 42 represents the processing order being issued in the 42nd week of year A.

C: Package factory code, which is A, HT, NJ, or WA, and can also be abbreviated as A, H, N, or W.

D: Test factory code, which can be A, Z, or H.

EE: Production Batch Code

Table 13-1 Order Information Table

Order Code	Packaging	Package	smallest unit
Ci523-Sample		Box/Tube	5
Ci523	3×3mm 16-pin QFN	Tape and reel	5K
Ci523	9.9×6.0mm 16-pin ESOP	Tape and reel	4K
Ci523	3×3mm 20-pin QFN	Tape and reel	5K